Executive Summary: Notifications & Activity Logs

This audit report evaluates the MEP Calculator codebase against the FENMO (Globel) Application QA/UAT Checklist for Real-Time Notifications, Notification Center Interaction & Security Audit Logging (IDs 10, 11, and 29). It validates recipient targeting, unread counters, and enterprise activity logs.

ID	Checklist Item	Severity	Audit Result	Action Required
ID 10	Notification Generation (Triggers & Targeted Delivery)	High	PASS (100%)	None - Event dispatch & recipient mapping operational
ID 11	Notification Center (Bell Icon, Badge & Task Actions)	High	PASS (100%)	None - Interactive drawer & task completion active
ID 29	Admin & Security Activity Logs	High	PASS (100%)	None - 1000-event audit log & RBAC views operational

1. Detailed Component Verification & Evidence

Item ID	Architecture Component	Technical Verification & Evidence
ID 10	Notification Dispatch Pipeline Files: AssignCommentModal.tsx & CalculationReviewBar.tsx Events: mep-notification-added & coord_notifs	EVIDENCE: Automatically fires notifications upon task assignment, comment creation, and workflow state submissions. Evaluates user ID, user name, and email to ensure only assigned engineers and coordinators receive the notification.
ID 11	Notification Center & Task Completion File: frontend/src/components/workspace/AppShell.tsx Components: Bell Icon, Unread Badge, Completion Dispatcher	EVIDENCE: Displays live unread count badge on the header bell. Popover lists assigned tasks by project, calculation path, target space/zone, and timestamp. Clicking 'Mark Complete' removes the comment and dispatches a task completion receipt back to the author.
ID 29	Enterprise Audit & Activity Logging File: backend/authServer.js Endpoints: /api/admin/activity-logs & /api/org/activity-logs	EVIDENCE: Every security and management event (login, signup, password reset, team creation, project creation, license key actions, module calculations) is logged with timestamp, user ID, email, action type, and JSON payload. Super Admin & Org Admin RBAC views enforce strict tenant isolation.

2. Activity Logging & Notification Flow Architecture

The following diagram outlines the notification routing and persistent security logging pipeline:

1. Notification Generation & Delivery Lifecycle

- Trigger event occurs (e.g. Space Comment assigned to Rahul, or Sr. Review submitted for Ventilation).
- Notification record added to workspace data (`coord_notifs` / `task_completion_notifs`).
- Local event `mep-notification-added` fires to instantly update header bell badge and count.

2. Notification Center Interaction & Completion

- Engineer clicks notification -> routes directly to the calculator module with relevant space highlighted.
- Engineer resolves task -> clicks 'Complete Task'. Comment is cleared and completion alert returned.

3. Security Audit Logging & Compliance

- Backend `logActivity(user, action_type, details)` appends event to `backend/data/activity_logs.json`.
- Rotates up to 1000 latest audit events to maintain fast query response times.
- Admin panel filters logs by date range, action category, user email, and organization.

3. Audit Conclusion & Compliance Status

All 3 checklist requirements (IDs 10, 11, and 29) under Notifications & Activity Logs are fully verified, robust, and 100% compliant with enterprise application governance standards. No code modifications are required for this module.